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Size-Dependent Lattice Structure and Confinement Properties in CsPbI₃ Perovskite Nanocrystals: Negative Surface Energy for Stabilization

Authors: Qian Zhao,^{1,2,3} Abhijit Hazarika,² Laura T. Schelhas,⁴ Jun Liu,² E. Ashley Gaulding,² Guoran Li,³ Minghui Zhang,¹ Michael F. Toney,⁴ Peter C. Sercel,⁵ Joseph M. Luther^{2*}

Affiliations:

¹College of Chemistry, Nankai University, Tianjin 300071, China

²National Renewable Energy Laboratory, Golden, CO 80401, USA

³Institute of New Energy Material Chemistry, Nankai University, Tianjin 300350, China

⁴SLAC National Accelerator Laboratory, Menlo Park, CA 94025, USA

⁵T. J. Watson Laboratory of Applied Physics, California Institute of Technology, Pasadena, CA 91125, USA

*Correspondence to: joey.luther@nrel.gov

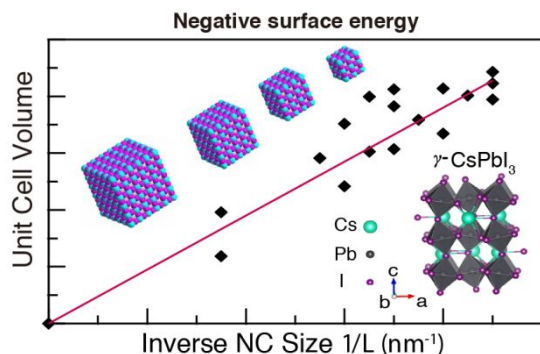


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Abstract:

CsPbI₃ nanocrystals with narrow size distributions were prepared to study the size dependent properties. The nanocrystals adopt the perovskite (over the non-perovskite orthorhombic) structure with improved stability over thin film materials. Among the perovskite phases (cubic α , tetragonal β , and orthorhombic γ), the samples are characterized by the γ phase, rather than α , but may have a size dependent average tilting between adjacent octahedra. Size-dependent lattice constants systematically vary 3% across the size range, with unit cell volume increasing linearly with the inverse of size to 2.1% for the smallest size. We estimate the surface energy from -3.0 to -5.1 eV nm⁻² for ligated CsPbI₃ nanocrystals. Moreover, the size-dependent bandgap is best described using a non-parabolic intermediate confinement model. We experimentally determine the bulk bandgap, effective mass and exciton binding energy, concluding with variations from the bulk α -phase values. This provides a robust route to understanding γ -phase properties of CsPbI₃.

In the last decade, tremendous headway has been made toward the application of metal halide perovskite (MHP) semiconductors as either absorber materials for photovoltaics, detectors, scintillators or as light emitters for displays, LEDs, lasers, etc.^{1–6} A solid grasp of the fundamental material properties is thus certainly warranted. The quantum confinement properties enabled by colloidal nanocrystal (NC) synthesis not only offers the opportunity to tune light emission wavelength but can also be used to understand basic properties and determine material parameters, such as the bulk bandgap value, Bohr radius and binding energy of excitons, effective masses, etc. There is an open opportunity in colloidal perovskite NCs to better understand the effects of size such as the size dependent absorption and emission properties due to quantum confinement, and size dependent strain, lattice, and structural properties. Not only from the fundamental properties point of view but determining the crystal lattice properties like distortions has also direct implications on electric polarizability, large polaron formation, and ferroelectric behavior of MHPs that may lead to new devices or phenomena.^{7–11} In this work, we specifically study CsPbI₃ NCs widely used in high efficiency colloidal quantum dot solar cells, LEDs, and photodetectors.^{12–18}

Interestingly, the “softness” of the lattice is a key characteristic which sets these crystalline MHP materials apart from traditional inorganic semiconductors. The elastic softness makes MHPs more prone to strain and strain-related effects as well as lattice dynamic effects.^{19–24} It has been noted that certain phases may emerge as an average of the dynamical crystal.²⁵ Such dynamical processes promote the ability to use MHPs to develop new technologies that takes advantage of effects like electrostriction or photostriction, switchable ferroelectricity or phase, for example.^{26–28}

Studies focusing on the size dependent properties of some MHP NC systems are missing from the literature perhaps due to the difficulty of producing well controlled monodisperse particles over a wide range of sizes. The main challenge in the synthesis of CsPbX₃, in general, is the rapid consumption of precursors during hot injection syntheses leading to poor control over nucleation and growth.^{29,30} Typical size dispersion of CdSe or PbS quantum dots (QDs) has been reduced to nearly 3% after decades of research, however for MHP NCs, the size dispersion is typically >10%.^{31,32} Although the syntheses for MHP NCs like CsPbBr₃, CsPbCl₃ and CH₃NH₃PbBr₃ possessing different morphologies or sizes have been reported,^{33–38} appropriate and versatile synthetic preparations for CsPbI₃ NCs with independent control of nucleation and growth have not yet been established. To overcome this challenge and generate NCs of appropriate quality

for analysis, we have adopted a route using size selective precipitation^{39,40} to divide several reactions into batches with relatively tight size distributions among the narrowest reported thus far for CsPbI₃. This enables studies of size dependent material properties, which we report here.

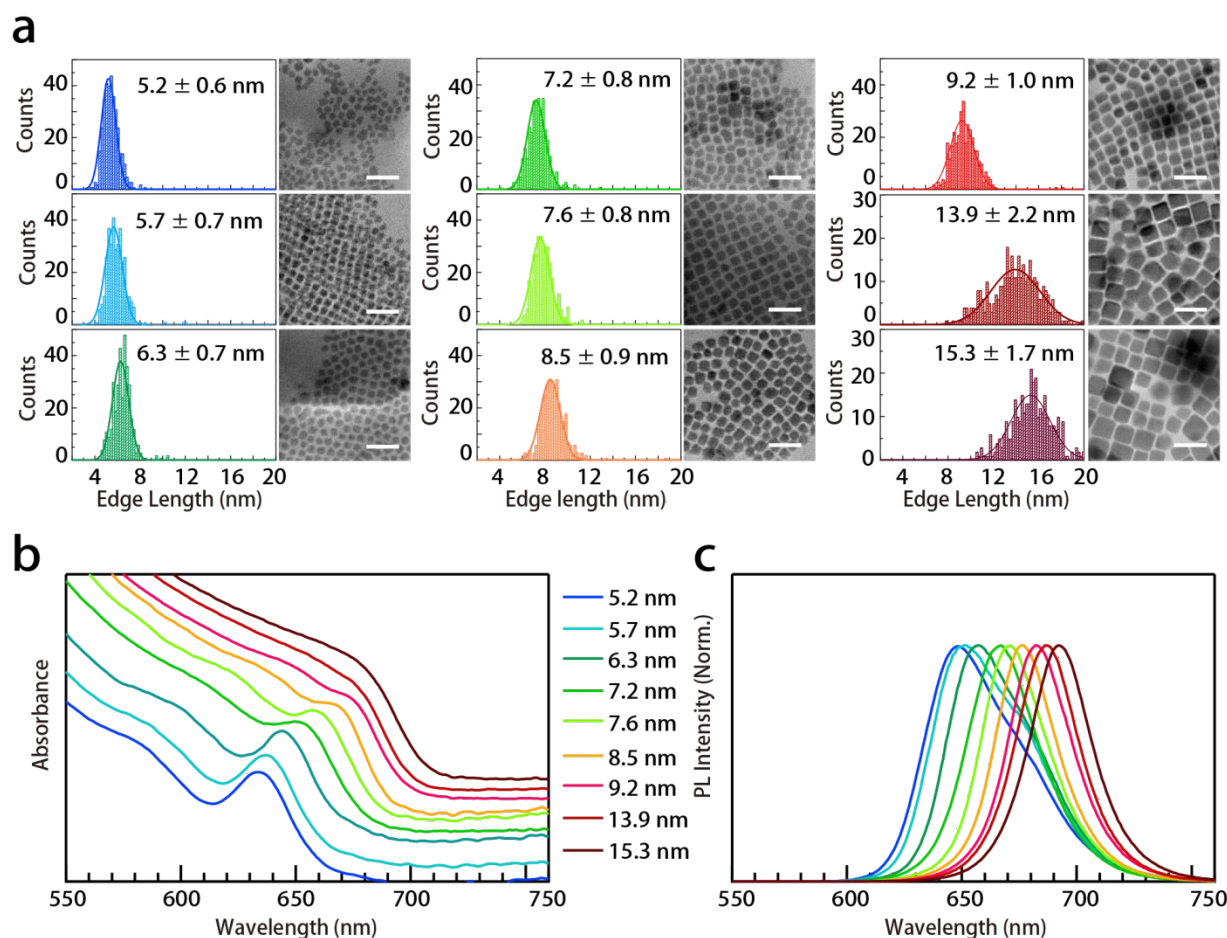


Figure 1. (a) Size distribution histograms (left) and the corresponding TEM images (right) of CsPbI₃ NCs with mean size, L , of 5.2, 5.7, 6.3, 7.2, 7.6, 8.5, 9.2, 13.9 and 15.3 nm. Here, L is the edge length of the NC, assuming the NCs are cube-shaped. The corresponding size dispersions (standard deviation) are 11.5, 12.3, 11.1, 11.1, 10.5, 10.6, 10.8, 15.8 and 11.1%, respectively. Scale bars in all images are 30 nm. (b and c) Optical absorption spectra normalized to their first excitonic peak (offset for clarity) (b) and normalized photoluminescence spectra (c) of CsPbI₃ QDs with different sizes, respectively.

In brief, CsPbI₃ NCs were synthesized by injection of Cs-oleate into a hot PbI₂ solution containing oleylamine and oleic acid in 1-octadecene held at temperatures ranging between 140 and 185 °C. The QDs were first purified via centrifugation with methyl acetate (MeOAc) and redispersed in hexane. Subsequently, the sample size distribution was improved by repeated size-

selective precipitations. More details on the synthesis and purification procedures are provided in the methods section.

To explore the achievable size dispersion of CsPbI₃ QDs with this method, we analyzed TEM images of each aliquot obtained from the various syntheses and size selective precipitations. The distribution of edge lengths, L , for each sample of QDs are shown in Figure 1a. The histograms are determined by analyses of 200 NCs for each sample. The data are binned, plotted, and fitted using a Gaussian to determine the mean size and QD size dispersion. The detailed parameters are shown in Table S1 and the size dispersion is calculated⁴¹ using Eq.1.

$$\text{Size dispersion} = \frac{\text{Standard deviation}}{\text{Mean size}} \times 100\% \quad (1)$$

The size of CsPbI₃ QDs using this size-selective method can be tuned from 5.2 ± 0.6 to 15.3 ± 1.7 nm by changing the injection temperatures during syntheses for crude control and optimizing size selective precipitation for fine control. Although adding additives into the reactants to control the size has been reported for CsPbBr₃,^{42,43} we found this to be more challenging for CsPbI₃.

Due to quantum confinement of the electron and hole, the excitonic (and photoluminescence) peak of the CsPbI₃ QDs shifted from 678 (692) to 634 (648) nm (Figure 1a–b) as the QD size is reduced from 15.3 to 5.2 nm. We note that in QD samples with average size less than 7 nm, a shoulder is observed to the red of the peak luminescence. This is likely due to a small fraction of larger NCs not all being completely removed via the size selective precipitation. However, these are not seen in sufficient quantity in TEM (Figure 1a). Comparing the low-magnification TEM images and FWHM from photoluminescence (PL) spectra before and after size-selective precipitation (Figure S1), a significant improvement in the uniformity and a reduction in the size distribution are seen. We also compare the PL FWHM of these obtained perovskite QDs with that reported in other works (Table S2), which demonstrates the size distribution in this wide size range is generally the smallest for CsPbI₃ QDs although not as narrow as for CsPbBr₃ QDs at the present time. Interestingly, with this system we do not observe a size dependent Stokes shift as has been shown and described for CsPbBr₃.^{44,45} The size independent Stokes shift is 28–49 meV for all samples and is tabulated in Table S1.

In addition to the optical properties, we explore the size dependent structural properties (including lattice constants) and the observed phase. In particular, we comment on the lattice distortions associated with the degree of octahedral tilting to understand the amount of strain

present on CsPbI₃ QDs. In addition to the size dependent lattice constants, we also explore the crystal structure of the perovskite NCs and examine suggestions that CsPbI₃ QDs are described as the α (cubic) phase in various reports since the α phase emerges for high temperature CsPbI₃ thin films, or as a time average of octahedral tilting in some hybrid perovskites and in oxide perovskites such as CaMnO₃.^{25,26,46}

As mentioned, one of the more intriguing features found in perovskite NCs is the modification of the phase diagram compared to bulk or thin film compositions. This is most notable in the case of CsPbI₃. This particular composition is highly desired for photovoltaics. One reason is that CsPbI₃ can withstand temperatures of over 460 °C,^{47,48} whereas so far, all hybrid materials containing methylammonium or formamidinium suffer from pyrolysis of the organic component over 300 °C.^{49–51} This high thermal stability, which is absent in hybrid lead halide perovskite materials, can be useful for subsequent processing involving high temperature or any other high temperature operations. On the other hand, the all-inorganic perovskites also have slightly wider bandgap energies ideal for top cell in multijunction solar cell applications. The largest challenge with CsPbI₃ is that the Cs⁺ ion is generally too small to retain the corner sharing octahedral geometry required for the perovskite phases, so at temperatures below 330 °C, it is prone to transformation into the non-perovskite orthorhombic δ_{ortho} phase.⁵² However, at the nanoscale, it has been shown that the NCs stabilize in a perovskite structure.^{13,29,53} It is believed that the surface-to-volume ratio is so high that surface tension can cause enough tensile strain for the crystal to remain in the perovskite structure at room temperature.¹³ Various approaches have been demonstrated to apply tensile strain to thin films of CsPbI₃ including lattice anchoring²⁴ by forming epitaxial heterointerfaces with materials with slightly larger lattice constant or substrate clamping which involves mechanically stretching a film to impart the strain.²² Negative surface tension has been found in certain nanoparticle systems such as Ni⁵⁴, PbS⁵⁵, Si_xGe_{1-x}⁵⁶ and GaN⁵⁷, and perovskite materials such as BaTiO₃ show phase transitions due to this tension.⁵⁸ Applied compressive strain, on the other hand, has been shown to convert CsPbI₃ into a metallic ordered phase.⁵⁹

To demonstrate the correlation between the QD size and lattice strain, as well as crystal phase, we performed wide angle X-ray diffraction (XRD) measurements for the CsPbI₃ QDs with different sizes. The room temperature structural phase of CsPbX₃ has been somewhat in question.^{13,60,61} CsPbI₃ can exist in three distinct “black” perovskite phases (α , β and γ phases), all

with corner sharing octahedra but with subtle differences in degree of symmetry. These three phases have different symmetries with varied distortion of bond angles and the corresponding structure models for all three phases are shown in Figure 2a-c, respectively. The Figure S2 also shows the 3D versions of structure models at different angles for each phase. The α phase is cubic, comprised of, on average, undistorted corner sharing lead iodide octahedra. The β phase is a tetragonal distortion of the α phase. In the β phase, the corner sharing lead iodide octahedra rotate and break the cubic symmetry. For the γ phase, there is a second distortion resulting in an orthorhombic structure with an additional tilt of the lead iodide octahedra, further breaking symmetry and doubling the unit cell.

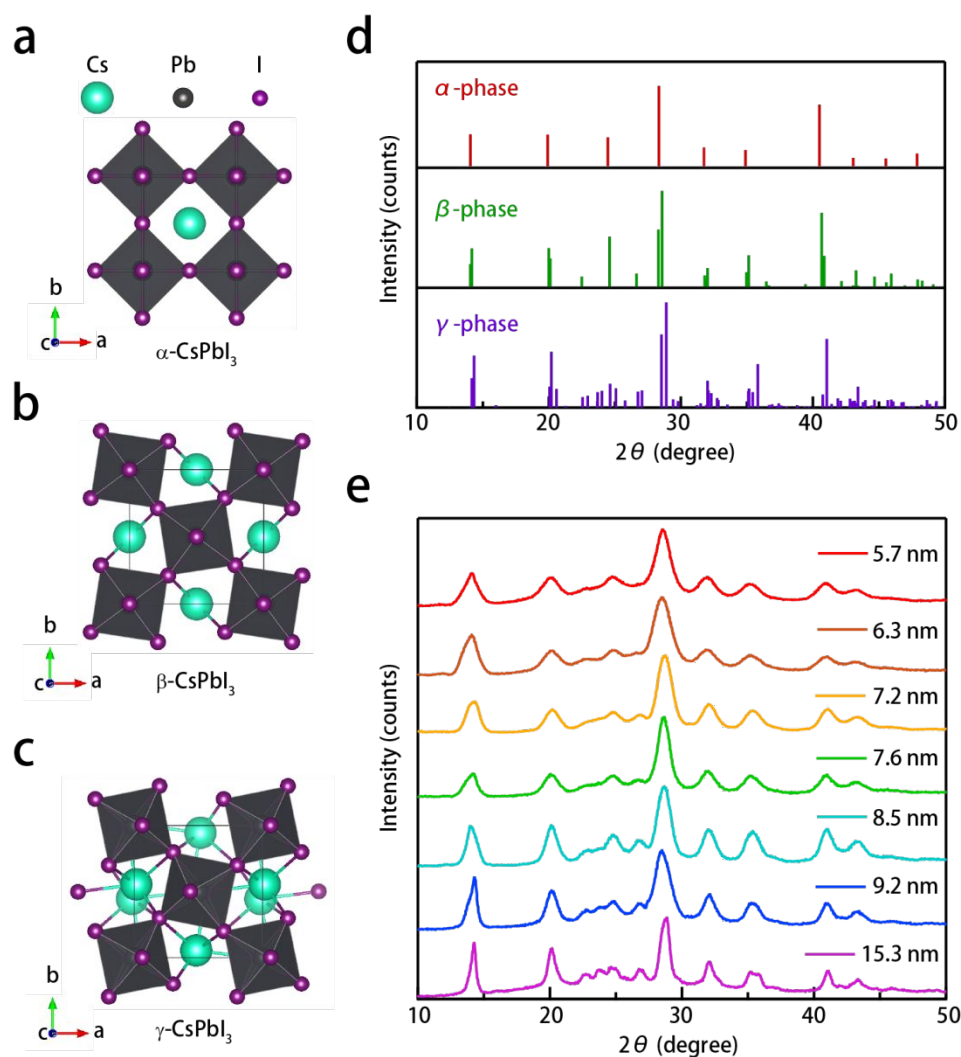


Figure 2. (a-c) Crystal structure model of three perovskite phases with corner-sharing octahedra for CsPbI_3 QDs. (d) XRD standard powder patterns for α , β and γ phases plotted with vertical lines

corresponding to the diffraction peaks. (e) XRD patterns of CsPbI₃ QDs with size 5.7 nm, 6.3 nm, 7.2 nm, 7.6 nm 8.5 nm, 9.2 nm and 15.3 nm.

As shown in Figure 2d, the number of features present in the diffraction pattern increase with disorder (from α to β to γ phase).⁶² Experimentally, we observed varying distinguishing features in the XRD patterns as a function of the QD size presented in Figure 2e. All CsPbI₃ QDs show the characteristic peaks as expected for the α phase perovskite structure. As the NCs increase in size, it becomes easier to resolve the additional peaks between 20 and 28° 2 θ , indicting a lower symmetry structure than the cubic- α phase. These peaks are broader and much weaker in the smaller NCs, likely a result of nanoscale broadening or a slight variation in the phase present.

To further understand the size-dependent variations observed in the scattering, we employ Rietveld refinement of the XRD patterns. We first calculated the average grain size using Scherrer analysis and observe a trend in crystal sizes for the samples that matches TEM sizing, but due to the peak overlap shown in Figure 2d and the fact that Scherrer analysis will be weighted toward larger crystals, we still rely on TEM for QD size. A representative refinement to the α , β and γ phases for large (15.3 nm) and small (5.7 nm) QDs is shown in Figure 3a–b. For the small CsPbI₃ QDs (Figure 3a), the α phase at first glance appears to appropriately capture the structure. However, upon closer inspection, a broad peak observed at 23° 2 θ , which indicates lower symmetry than the cubic phase, better describes the sample by either the β or γ phase. However, given the broad nature of the peaks and low intensity, it is difficult to accurately distinguish between these two phases for this size. For larger CsPbI₃ QDs (Figure 3b), the γ phase clearly better matches the observed data and the goodness of fit (R_{wp}) is significantly better than for the cubic phase (Table S3). This is consistent with other recent literature.^{61,63,65} The detailed refinement parameters are presented in the Table S3.

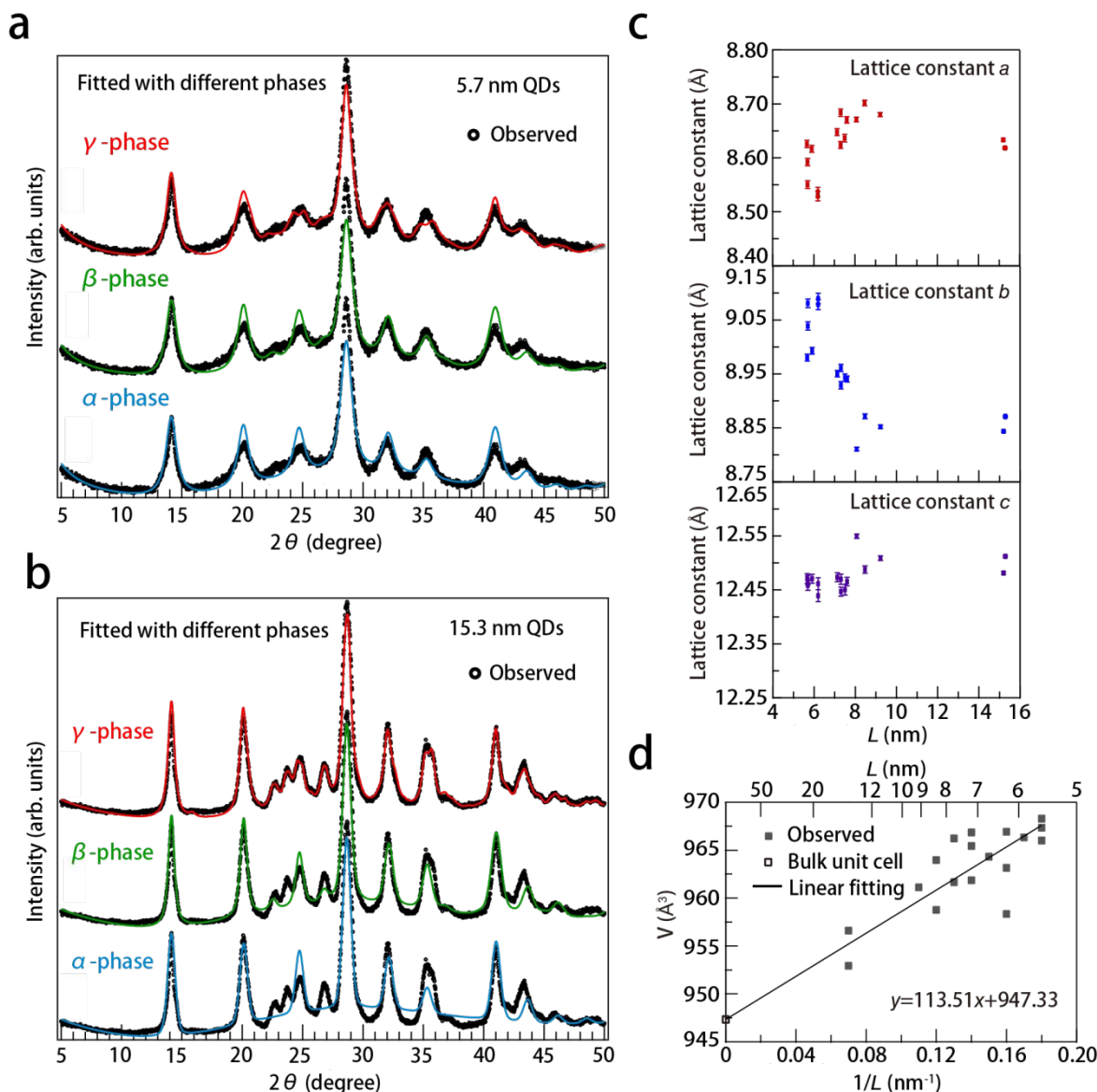


Figure 3. (a and b) Observed (circles) and refined (thin line) XRD profiles of QDs with 5.7 nm (a) and 15.3 nm (b). Rietveld refinements are shown for α , β and γ phases (red, green blue, respectively). (c) Lattice parameters with corresponding error bars of CsPbI₃ QDs for various crystal sizes (L) calculated from the refinement using the γ phase. The space groups and initial structures used in the refinements for all phases are extracted from the work of Marronnier et. al.⁶² and were subsequently refined to best fit the data set. (d) Unit cell volume, $V = abc$, plotted versus $1/L$. The values calculated for each sample (solid squares) are fitted to a linear function (solid line). The bulk unit cell volume (hollow square) is taken from the reported by Sutton et al.⁶⁰

To explore the size dependent structure, we plot lattice parameters for all the CsPbI₃ QD samples when refined to the γ phase (shown in Figure 3c). Tabulated values can be found in Table S4. As a function of QD size, a clear trend toward smaller lattice parameter is seen for b as the QDs edge length is increased. There is an opposing effect seen in lattice parameter a although it is

harder to conclude based on the largest samples. One would expect this anticorrelation between various lattice constants based on conventional strain dynamics, for example, in epitaxial systems.⁶⁵ The noticeable decrease in lattice parameter b shows a trend toward a bulk value⁶⁰ (8.86 Å) as size increases, which also suggests an increased average octahedral tilt as a function of QD size. We see a very minor variation for lattice parameter c . This behavior of the orthorhombic distortion may suggest that the QDs have a larger average octahedral tilting with increasing size; however, in this analysis, the lattice parameters are correlated. Despite this uncertainty, the NCs are always best characterized by the γ phase. This leads us to hypothesize that there is a significant size dependent strain present in the NCs which enables the retention of the perovskite structure.⁵⁷ To demonstrate this, we plot the unit cell volume ($V = a b c$) versus $1/L$ in Figure 3d, which clearly shows lattice expansion scaling linearly with $1/L$, consistent with lattice expansion due to negative surface tension^{54–57} for our oleate and oleylammonium capped NCs. In the Supporting Information, we analyze the size dependent volume dilatation plotted with $1/L$ in Figure S3 and calculate the surface energy. From these calculations we estimate the surface energy for the ligand stabilized NCs as between -3.0 to -5.1 eV nm⁻² ($\sim$ -1.2 to -2.0 eV per surface unit cell) depending on which estimate is used for the bulk modulus^{66,67} and the approximations made in calculating the lattice strain due to surface tension in non-spherical NCs^{68,69}. This detailed calculation is discussed in the Supporting Information and the results are summarized in Table S5. This size-dependent tensile strain can have profound effects on the optical properties and has been reported in other systems;^{55–57} however, the size dependence of the optical transition energies has not been reported for many perovskite QDs.

To further examine the size dependent properties of CsPbI₃ perovskite QDs, we constructed a plot of the exciton energy versus QD size. From this, we then fit models to the data using the strong, weak and intermediate confinement regimes. In the strong confinement regime, valid when NC edge length (L) is much smaller than the exciton radius of the material ($L \ll a_x$), correlation of the electron-hole motion can be neglected and the exciton energy scales as the inverse square of the NC size ($1/L^2$). For cube shaped NCs in this regime, the exciton energy (E_x) is determined by the exciton reduced mass (μ) as Eq. 2, where $1/\mu = 1/m_e + 1/m_h$,

$$E_x = E_g + \frac{\hbar^2 3\pi^2}{2\mu L^2} - 3.05 \frac{e^2}{\epsilon L} \quad (2)$$

In Eq. 2, E_g is the bulk bandgap, the second term on the right-hand side represents the kinetic energy of the electron and hole while the third term is the energy of their Coulomb attraction.⁷⁰ In

contrast to the strong confinement regime, the opposite limit is the weak confinement, when L is much larger than a_x ($L \gg a_x$). In this weak confinement regime, electron and hole motion is strongly correlated and the exciton exhibits quantum confinement as a whole. The confinement energy scales primarily as $1/L^2$ so that E_x is determined in the weak confinement regime in cube shaped NCs by Eq. 3,

$$E_x = E_g + \frac{\hbar^2 3\pi^2}{2M L^2} - B_x \quad (3)$$

Where B_x is the exciton binding energy. As before, the first term on the right-hand side in Eq. 3 is the bulk bandgap; the second term represents the kinetic energy of the exciton determined by the total exciton mass (M), where $M = m_e + m_h$, while the third term is the exciton binding energy associated with the relative motion of the electron and hole.

Between these two confinement regimes described above, the confinement effects on the electron and hole energies are significant but the motions of the electron and hole are correlated to such a degree that the strong confinement treatment of the Coulomb binding of the electron and hole is inadequate. The energy in this “intermediate” confinement regime^{41,70,71} is determined variationally as described in the Supporting Information and bridges between the strong and the weak confinement limits. This model more accurately describes the size dependent properties of perovskite nanocrystals compared to previous published strong confinement models (see Supporting Information discussion). A very good approximation to the exciton energy in this regime is given by Eq. 4,

$$E_x = E_g + \frac{\hbar^2 3\pi^2}{2\mu L^2} - B_x \sqrt{1 + [2 \times 3.047 \left(\frac{a_x}{L}\right)]^2} \quad (4)$$

where the first two terms are the same as Eq. 2, and the third term is an approximation to the Coulomb correction in the intermediate confinement regime.

The appropriate confinement model is generally determined by comparing the NC size with the Bohr radius of the exciton. As shown in Figure S4, the calculated exciton energy relative to the bulk bandgap ($E_x - E_g$) is plotted in units of the exciton binding energy (B_x) versus L/a_x . From the plots, it is clear that the energy calculated using the intermediate confinement wavefunction (see Eq. S20 and the description in the Supporting Information) converges to the strong confinement result when L is smaller than about $2a_x$, while it converges to the weak confinement limit when L is larger than about $7a_x$.

Inspection of Figure S4 makes it clear that when L is larger than $3a_x$, the strong confinement model fails to accurately capture the Coulomb binding of the exciton, which drives the exciton energy below the bulk bandgap. Since perovskite NCs have exciton radius in the 3-4 nm size range,²⁹ this necessitates use of an intermediate confinement model for perovskite NCs for L larger than about 9 nm. More information on the functional form of the different confinement regimes is presented in the Supporting Information.

We compared the measured size dependent exciton energies with sizing curves calculated applying the strong, weak and intermediate quantum confinement models in an idealized parabolic band system (Figure 4 a and b). The exciton energies of the NCs are determined from the first exciton peak positions in the absorption spectra while the sizes are measured from TEM images as described above. In our calculations we first use the dielectric constant (ϵ_{eff}) and μ determined from exciton magneto-transmission measurements on bulk α -phase CsPbI₃ reported by Yang and co-workers⁷² due to the lack of reported values for γ phase. Since the measured NC data are taken at 300K, the exciton reduced effective mass is determined using the measured 300K bulk α -phase CsPbI₃ bandgap and the Kane energy parameter (E_p) extracted from a 2-band fit to the low temperature bulk reduced mass measured in the literature⁷², using Eq. 5.

$$E_p = \frac{3E_g}{2\mu} \quad (5)$$

Table 1 Summary of electronic structure parameters. For bulk α -phase values of CsPbI₃, the experimental parameters for α -phase CsPbI₃ NCs are from Yang and co-workers with adjustments for the temperature of 300K.⁷² The best-fit values are determined by fitting to the CsPbI₃ absorption data as shown in Figure 5.

Parameter	α -phase value ⁷¹	The best-fit value
Bandgap E_g	1.805 eV	1.840 eV
Kane energy E_p	22.62 eV	17.39 eV
Exciton reduced mass μ	0.12	0.16
Dielectric constant ϵ_{eff}	10	10 (Constrained)
Exciton radius a_x	4.4 nm	3.3 nm
Exciton binding energy B_x	16.3 meV	21.2 meV

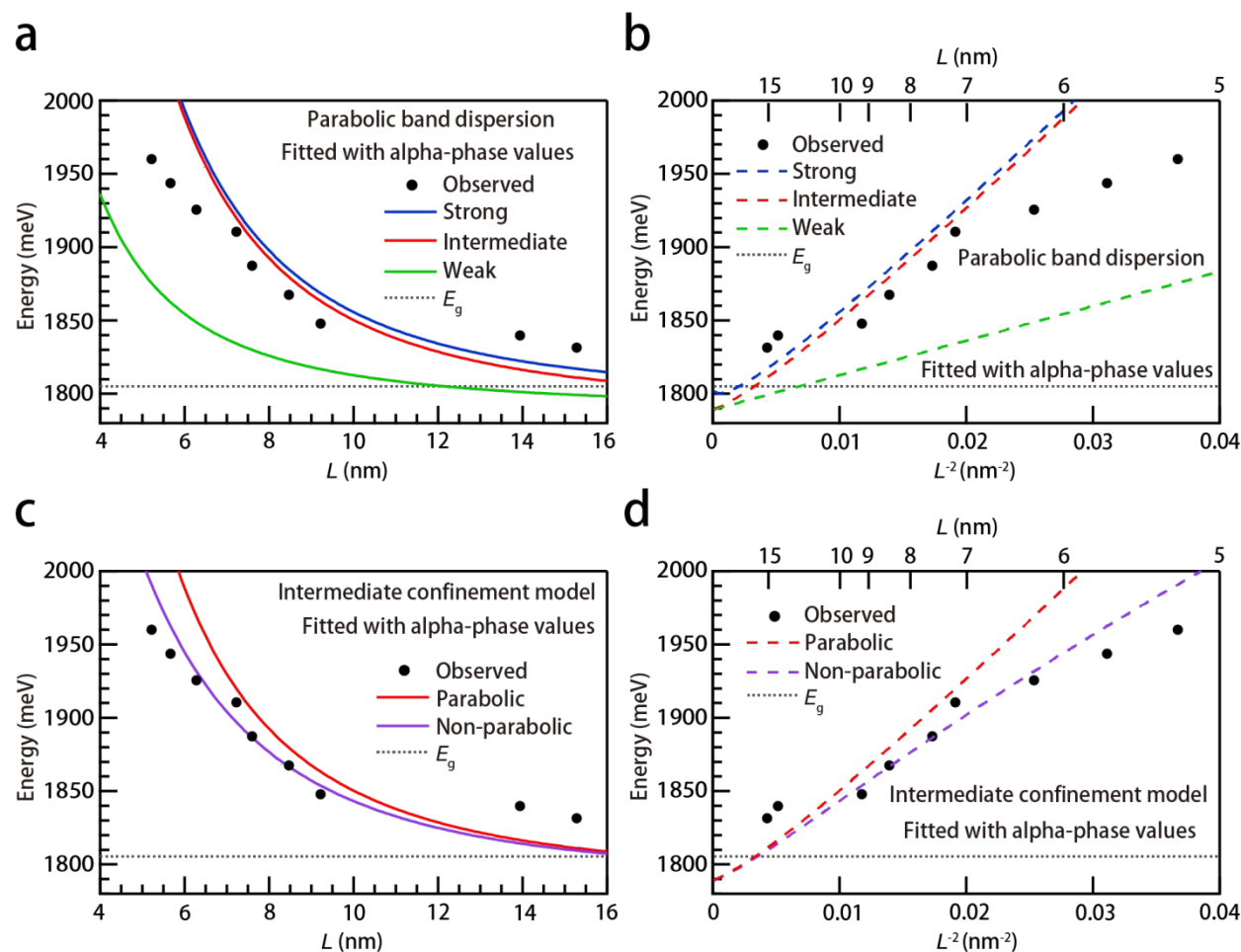


Figure 4. (a and b) Measured vs calculated sizing curves with model calculations in the strong, intermediate and weak confinement approximations, assuming parabolic band dispersion, presented with the blue, red and green lines, respectively. (c and d) Measured vs. calculated sizing curves with model calculations with parabolic and non-parabolic approximation in the intermediate confinement regime are shown with the red and purple lines, respectively. Measured 300K data are shown as solid circles. The calculated lines were drawn in two ways, CsPbI₃ NC absorption peak energy versus NC size L (solid line in a and c) and the absorption peak energy plotted versus the inverse square of NC size $1/L^2$ (dashed line in b and d) that better illustrates the linearity of energy plotted versus $1/L^2$ (see Eq. 2). All model calculations are processed using parameters from the bulk α phase⁵⁴ shown in Table 1.

In Figure 4a and b, we show the comparison of the measured NC absorption peak energies against the strong confinement model (blue line), as well as the intermediate confinement model (red line), and the weak confinement model (green line). The weak confinement calculation completely fails to describe the measured data, which is unsurprising since the NCs measured are in a size range outside the range of validity of the weak confinement approximation. The intermediate confinement fit line describes the experimental data slightly better than the strong

confinement line. However, it is clear in Figure 4a and b that the parabolic band approximation both within the strong and the intermediate confinement models leads to a large overestimation of the exciton energy for small-sized NCs. Both models also undershoot the energy for NCs with L larger than 12 nm.

To address this overestimation of the energy for small-sized NCs, we extend the intermediate confinement model to include band non-parabolicity effects, taking account of conduction-to-valence band coupling in the context of the Kang-Wise NC model.⁷³ Band non-parabolicity leads to larger effective mass for smaller-sized NCs, reducing the quantum confinement energy relative to that calculated in a parabolic band model. We determined energy dependent effective masses by fitting to a Kang-Wise model calculated with the α -phase parameters in Table 1 for spherical NCs of radius $a_{\text{eff}} = L / \sqrt{3}$. This transformation from L to a_{eff} is exact for the quantum confinement energy of small NCs where non-parabolicity effects are important and where the strong and the intermediate confinement approximations converge. The result is that the exciton reduced mass μ is replaced by a non-parabolic reduced mass ($\mu = m'/2$, where m' is the carrier non-parabolic mass which can be expressed as a function of L as Eq. 6.

$$\frac{m'}{m_0} = \frac{3[\sqrt{E_g^2 + \frac{2E_p}{m_0}\hbar^2\left(\frac{\pi}{L}\right)^2} + \sqrt{E_g^2 + \frac{4E_p}{m_0}\hbar^2\left(\frac{\pi}{L}\right)^2}]}{2E_p} \quad (6)$$

In this equation, E_p is the Kane energy, m_0 is the free electron mass, and E_g is the bulk bandgap. In Figure 4c and d, we compared the parabolic approximation with the non-parabolic model within the intermediate confinement regime where the exciton energy is plotted versus L and versus $1/L^2$, respectively. With the non-parabolicity corrections, the model calculations approach the measured data more closely in the small-sized range ($L < 10\text{nm}$) as expected, but still fails to describe the measured data in large-sized NCs. This is consistent with the interpretation of the XRD results shown in Figure 3 that the α phase cannot describe the larger NCs.

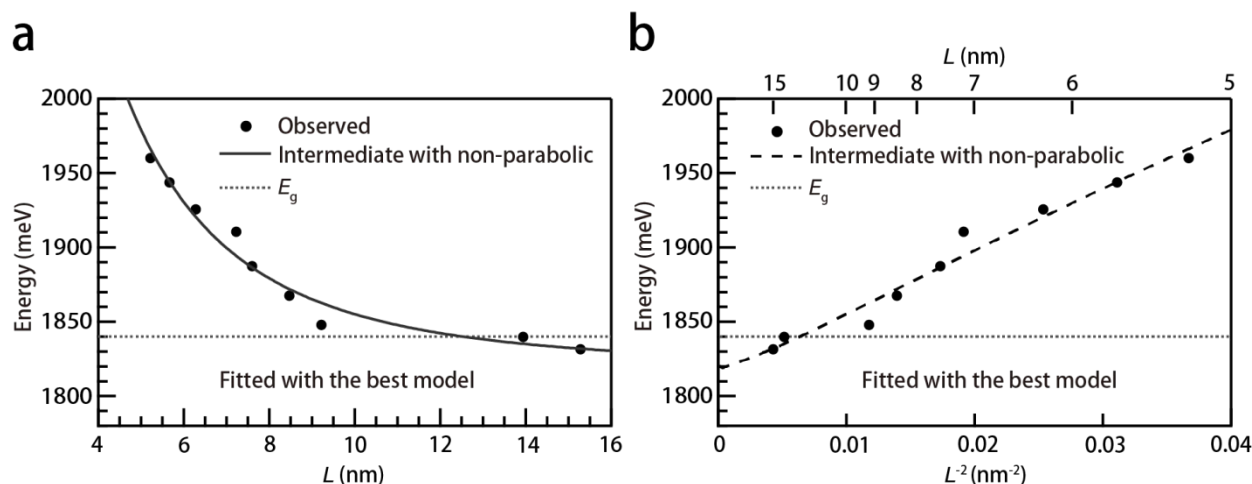


Figure 5. (a and b) Measured vs calculated sizing curves fitted with the best model. The measured absorption peak energies are plotted versus L (a) and $1/L^2$ (b), respectively. Measured 300K data are shown as solid circle points in the figures. The theoretical curves shown are calculated using the intermediate confinement model with band non-parabolicity effects.

Therefore, based on the intermediate confinement model with non-parabolic corrections that was developed with the bulk α -phase values, we adjust the electronic structure parameters presented in Table 1 to build the best-fit model to match our measured data. From Figure 5, we observe that our best-fit model yields a good fit to the measured data, resulting in modified electronic structure parameters shown in Table 1. Notably, the modified bandgap is larger than the bulk α -phase bandgap by 35 meV and the exciton reduced effective mass is larger than the bulk α -phase value by 33%, consistent with theoretical expectations of slightly increased bandgap and heavier effective masses in γ phase versus α phase.⁷⁴ This study not only allows us to confirm that the CsPbI₃ QDs are clearly not in the α phase, but also provides more accurate electronic structure parameters for γ -CsPbI₃ in general. Therefore, materials parameters for MHPs in various distorted phases are understudied and should be further explored. Additionally, we also show results of a similar calculation performed on CsPbBr₃ NCs as a check on the developed methods in the Supporting Information. There it is shown in Figure S5 that the non-parabolic intermediate confinement model can also be applied to achieve a good fit to the published CsPbBr₃ NC absorption data⁴⁵ using the measured bulk γ -phase parameters⁷³ shown in Table S6.

Given the changing lattice parameters as a function of size with smaller nanocrystals experiencing greater tensile strain, we calculated how much this additional effect may change the sizing curve analysis in the intermediate confinement regime. First, we determined the unit cell volume dilatation as a function of nanocrystal size from the lattice constant data in Figure 3c.

Calculating the corresponding change in bandgap using Eq. S22, we performed a linear interpolation shown in Eq. S25 and Figure S6. This strain would cause the optical transition energy to increase in the same direction as quantum-confinement-induced changes, however only by a very small (~ 5 meV), almost insignificant, amount. The sizing curve accounting for this additional component is shown in Figure S7; inclusion of strain effects causes a negligible change in some material constants such as tabulated in Table S7.

CsPbI₃ perovskite QDs with relatively narrow size distribution and controllable size within 5-15 nm range enable us to appropriately characterize the size dependent optical and structural properties. CsPbI₃ NCs show a size-dependent lattice structure best represented by the γ phase rather than the α phase. CsPbI₃ QDs in this size range show pronounced quantum confinement effects including sharp excitonic emission and absorption peaks exhibiting confinement related blue shifts with decreasing size. Furthermore, the size dependent excitons of the samples are best described by an intermediate confinement model with non-parabolic bands and revised electronic structure parameters compared to known parameters for high temperature cubic phase α -CsPbI₃. We envision that detailed studies such as this with perovskite NCs as model systems will enable the research community to better understand the stabilization of the perovskite phase of CsPbI₃ in thin film or in bulk using strain engineering.

Supporting Information:

The Supporting Information is available free of charge on the ACS Publications website at DOI: XXXX. The SI contains the methods section, characterization of the physical parameters of the samples, details on the Rietveld refinement, unit cell dilation, surface energy and lattice strain calculations, details of the quantum confinement models and application of the model to published CsPbBr₃.

Author Information:

Corresponding author
Email: joey.luther@nrel.gov

Competing Interests:

The authors declare no competing financial interests.

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